

Software Process And Models (SPAM)

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Abstract:

Software Engineering is more than just code; it requires an organized plan/process in order to be able to plan, develop, test, and maintain software systems. This project mentions different kinds of software development processes and explains why companies might choose one process or approach over another for their needs. The main models that are discussed are Waterfall, incremental development, integration and configuration, plan-driven development, and Agile development. I also go into four fundamental software process activities: specification, design and implementation, validation, and evolution.

To have a better understanding of how these approaches are used, I bring up two real-world applications that were researched: EpicCare Electronic Health Record (EHR), developed by Epic Systems, and League of Legends (LoL), developed by Riot Games. EpicCare is a healthcare software that requires careful planning, extensive testing, and reliable performance to support the care of patients. This system follows documented practices that suggest characteristics of a hybrid development approach. League of Legends, on the other hand, has frequent updates, automated testing, continuous integration, and player feedback, which suggest it uses Agile and incremental development for its system because of its gaming system and the players who play it, based on what I've researched from the information they've posted.

Based on my research, both of these systems require thorough planning, testing, etc. in order to have a functioning system. Their development priorities are different because they serve different users and have different necessities. Overall, this project demonstrates that each system requires a different process to build the system.

Technologies Used:

Problem / Motivation:

Software development is more than just writing code. Companies need organized processes in order to ensure that their software is reliable and efficient to meet user expectations. Choosing the wrong development approach could lead to delays, increased costs, and other problems. EpicCare EHR and LoL are good examples since they both serve different purposes, which allows us to compare two systems and how software engineers have to adapt to different systems and industries.

Software Process Concepts:

Software engineering includes four fundamental concepts: specification, design and implementation, validation, and evolution. There are also different process models that organize these concepts a little differently. (Sommerville, 2015).

Waterfall: follows structured phases.

Incremental and Agile: Allow for frequent updates and feedback.

Integration and Configuration: Focus on reusing existing software components.

Hybrid: Combines characteristics of multiple methods to meet specific project requirements (Sommerville, 2015).

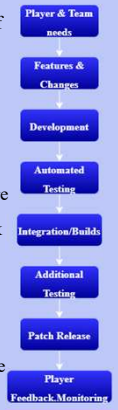
Real-World Application 1:

EpicCare is an electronic health record system that was developed by Epic Systems and is used by multiple healthcare organizations that manage patient information to help doctors monitor their patients and care plans. Its documented practices include, but are not limited to, workflows, software testing, system integration, and customer-specific implementation. EpicCare has multiple characteristics of a hybrid kind of system; it combines careful planning and validation with continuous improvements happening. Testing, controlled deployment, and maintenance are very important because healthcare organization software have to be reliable and must protect their sensitive data on patients. (Epic Systems Corporation, n.d.).



Real-World Application 2:

League of Legends is an online multiplayer game developed by Riot Games. The development of the game involves continuous integration, automated and some manual testing, frequent software patches, and player feedback. All of these characteristics point to agile and incremental development since new features are often released pretty regularly. Player feedback and ongoing monitoring help guide future updates, allowing Riot to improve gameplay, correct errors, and maintain the game's performance to give players the best experience possible (Riot Games, 2016).



EpicCare VS. LoL:

EpicCare and LoL both require organized development that involves a list of things like testing, deployment, and maintenance. But they both have completely different priorities. EpicCare has more clinical requirements, reliability, security, extensive validation, and controlled implementation. On the other hand, LoL focuses heavily on gameplay requirements, automated testing, frequent patches, and player feedback. EpicCare showcases a hybrid kind of development characteristics, while LoL demonstrates Agile and Incremental development characteristics.

Key Findings:

Comparing the two programs/systems, though very different, demonstrates that there's no single kind of development process that's appropriate for every single kind of software. Healthcare software like EpicCare requires multiple care considerations like validation and controlled implementation, because of any errors that might affect patient care if it's not programmed correctly. Then there's gaming software like League of Legends needs to have things like frequent updates and flexibility because of player requirements and expectations the are constantly changing. When it comes to a successful software system, you have to pick a process that works with what you need and key things like quality, cost, risk, and customer needs.

Future Considerations:

New technology that's coming out, like AI, automation, cloud computing, and DevOps, continues to influence software development, especially the more they evolve. For example, a system like EpicCare, AI-assisted testing, and improved cybersecurity tools might be able to help programmers to find errors and further ensure the patient information. While LoL, things like automated testing, cloud infrastructure, and AI-assisted development help programmers and developers to find bugs and deliver more efficient updates for players. Both systems have to adapt as technology evolves while also maintaining their system requirements. This doesn't just go for these two systems but for all of them.

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